

pH MEASUREMENT In Hydroponics System

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The base setup for this system is located in a room with approximate dimensions of 9.29 square metre. The room is non-air conditioned and does not receive any direct sunlight. The proposed setup aims to evaluate the feasibility of utilising an electrode-based E201 BNC pH measurement sensor for assessing the pH of the water before and after the addition of liquid nutrients.

Fig. 1 illustrates the authors' pro-

totype showcasing various aspects of the setup and its installation. Monitoring the pH value is crucial for maintaining optimal conditions for plant growth. The ESP32 is employed in deep sleep mode, and with the assistance of a touch sensor, the pH of the hydroponic solution can be measured with a simple touch. This touch sensor acts as an interrupt, triggering an external input for the ESP32, thereby awakening it from deep sleep mode. Fig. 2 pro-

vides a block diagram representation of the entire setup.

pH reflects the hydrogen potential in a solution, providing a numerical value that determines its acidity or alkalinity. The pH scale ranges from 0 to 14. A pH value below 7 indicates acidity, while a value above 7 indicates alkalinity. pH serves as a vital chemical indicator for hydroponic setups, reflecting changes in water due to the addition, absorption, and bal-

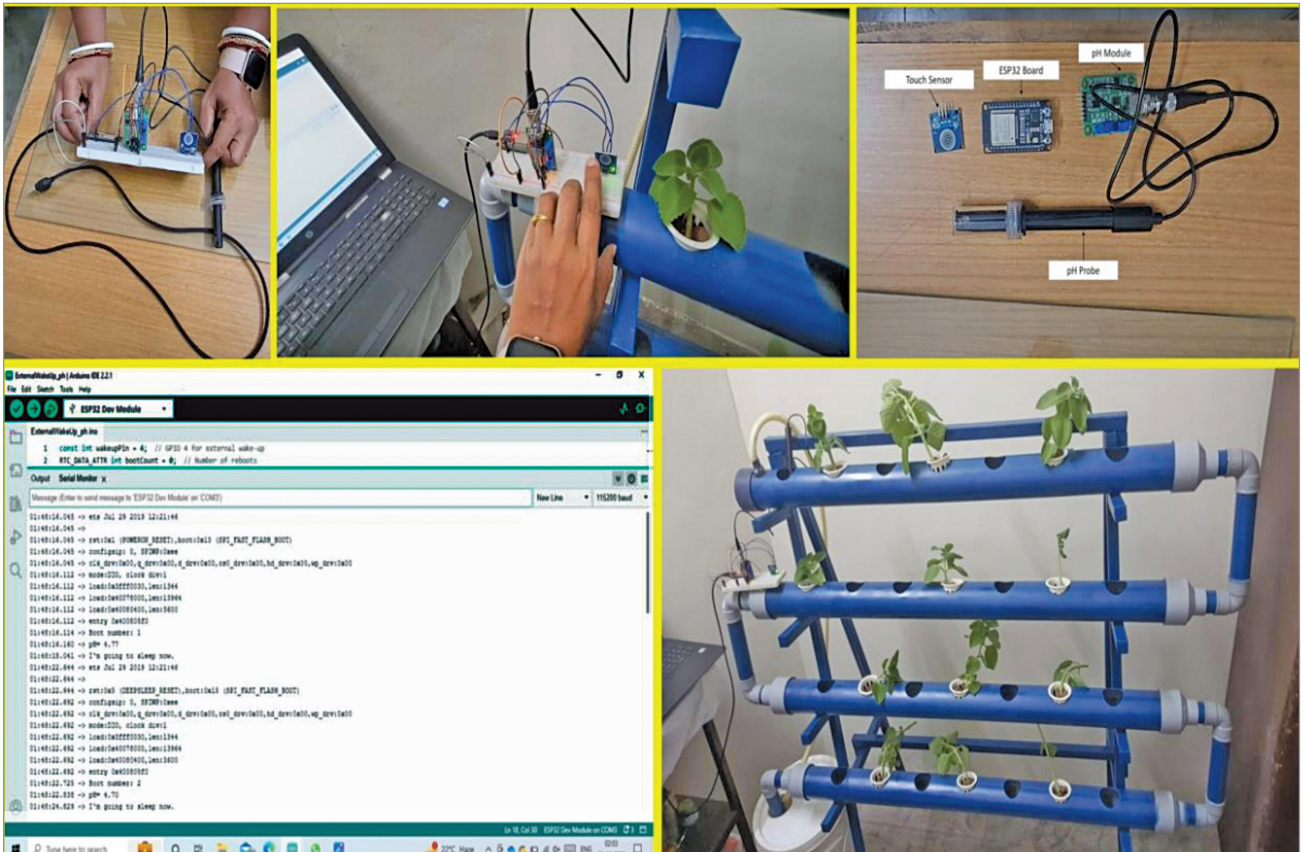


Fig. 1: Authors' prototype of the proposed setup for pH measurement